Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	08 Feb 2026
Team ID	LTVIP2026TMIDS77563
Project Name	Gemini Financial Decoder
Maximum Marks	4 Marks

Technical Architecture:

The project follows a Client–AI Architecture Model consisting of five layers:

1. User Interface Layer (Streamlit Web Application)
2. Application Logic Layer (Python + Pandas)
3. AI Processing Layer (Google Gemini API)
4. Output Rendering Layer (Charts + Financial Insights)
5. Deployment Layer (Local System / Streamlit Cloud

User
1. User Interface Layer
Streamlit Web App
• Upload CSV/XLSX financial files
• Display financial summaries
• Render charts and tables
2. Application Logic Layer
Python Backend
• File validation
• Data loading with Pandas
• Data preprocessing
3. Data Processing Layer
Financial Data Processing
• Convert DataFrame → Dictionary
• Extract numeric columns
• Prepare structured input for LLM
4. AI Analysis Layer
Google Gemini 2.5 Flash
• Balance Sheet analysis
• Profit & Loss insights
• Cash Flow evaluation
• Financial reasoning
5. Output & Visualization L
Insights & Visualization
• AI-generated summaries
• Financial ratio insights
• Streamlit charts
• Data tables

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based interface for uploading financial files and viewing insights	Streamlit
2.	Application Logic-1	Financial data reading and preprocessing	Python, Pandas
3.	Application Logic-2	Prompt construction for financial analysis	Python
4.	Application Logic-3	Session state handling and user interaction management	Streamlit Session State
5.	File Storage	Temporary storage of uploaded CSV/XLSX files during analysis	Local Runtime Memory
6.	External API-1	AI processing for financial analysis and summary generation	Google Gemini API
7.	Machine Learning Model	Generative AI model used for financial insight generation	Gemini Flash Model

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Application built using open-source libraries	Python, Streamlit, Pandas
2.	Security Implementations	API keys stored securely using environment variables	.env file, OS Environment Variables
3.	Scalable Architecture	Modular architecture separating UI, processing, and AI layers	API-based AI integration
4.	Availability	Web-based application accessible through browser	Streamlit Cloud / Local Deployment
5.	Performance	AI-generated financial summaries produced within a few seconds depending on data size	Gemini Flash API